



CSC236: Object Oriented Programming (2)

Final Project Report Template

Project Group Number (As in Blackboard): 1

Section Number: FS2

Project Title: Cinema Booking



Members:

	Name	ID	Role Participated in	Sign
1	(Leader)	2240001482	Rayana Alshehri	
2		2240006768	Reema Zeyad AlMulla	
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4		2240002658	Fatimah Saad Alghamdi	
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7		2240001878	Miad Altammar	



1. Checklist Of Requirements Fulfilled In The Project:

Function / Application Aspect	Classes Management	Data Export/import	Professional Appearance	Error Handling	Data Access Tools and Techniques
Book a Ticket (Select Movie, Time, Seat)	✓	✓	✓	✓	✓
Select Seats (Seat Availability Check)	✓	✓	✓	✓	✓
Customer Management (Add/Manage Customers)	✓	✓	✓	✓	✓
Staff Management (Manage Movies & Schedules)	✓	✓	✓	✓	✓
View Available Movies (for Customers)	✓	✓	✓	✓	✓
Movie Management (Add, Edit, Delete Movies)	✓	✓		✓	✓
Apply Special Offers & Discounts	✓	✓		✓	✓
Process Payments (Multiple Methods)	✓	✓	✓	✓	✓
Screen Management (Regular, IMAX, 4K)	✓	✓	✓	✓	✓
Food & Beverage Ordering System	✓	✓	✓	✓	✓
User Profiles & Loyalty Program	✓		✓	✓	✓
Refund & Cancellation Management	✓	✓		✓	✓

2. Difficulties & How Problems Were Solved:

One of the main challenges we faced during the project was agreeing on a unified design for the interfaces. Each team member initially had her own creative vision, which made it difficult to maintain a consistent theme and user experience across the application. Additionally, dividing the work equally while aligning with everyone's ideas required a great deal of communication and compromise. Another significant difficulty was connecting the different interfaces together, especially integrating them smoothly with the database. This process involved both frontend and backend coordination, which proved to be technically demanding. To overcome these challenges, we held regular team meetings on Microsoft Teams to discuss issues, align our design approach, and ensure everyone was on the same page. We also made use of online resources such as YouTube tutorials and sought advice from more experienced peers to find solutions. These collaborative efforts greatly contributed to resolving our technical and design-related difficulties and helped us move forward effectively as a team.



3. How Were the Tasks Distributed Equally Between Members?

When we first chose our group project, we had a discussion to ensure the tasks were fairly distributed. We agreed that each member would be responsible for designing and implementing their own GUI interface. This included creating the visual components, user interactions, and relevant functionality. For larger GUI interfaces, we decided to split the work between two members to ensure fairness and a balanced workload. In addition to the interfaces, each member also worked on their own set of classes related to the functionality of their specific part of the project. By dividing the tasks in this way, everyone had a clear responsibility and contributed equally to the project, resulting in a balanced workload for all members.



4. General Comments:

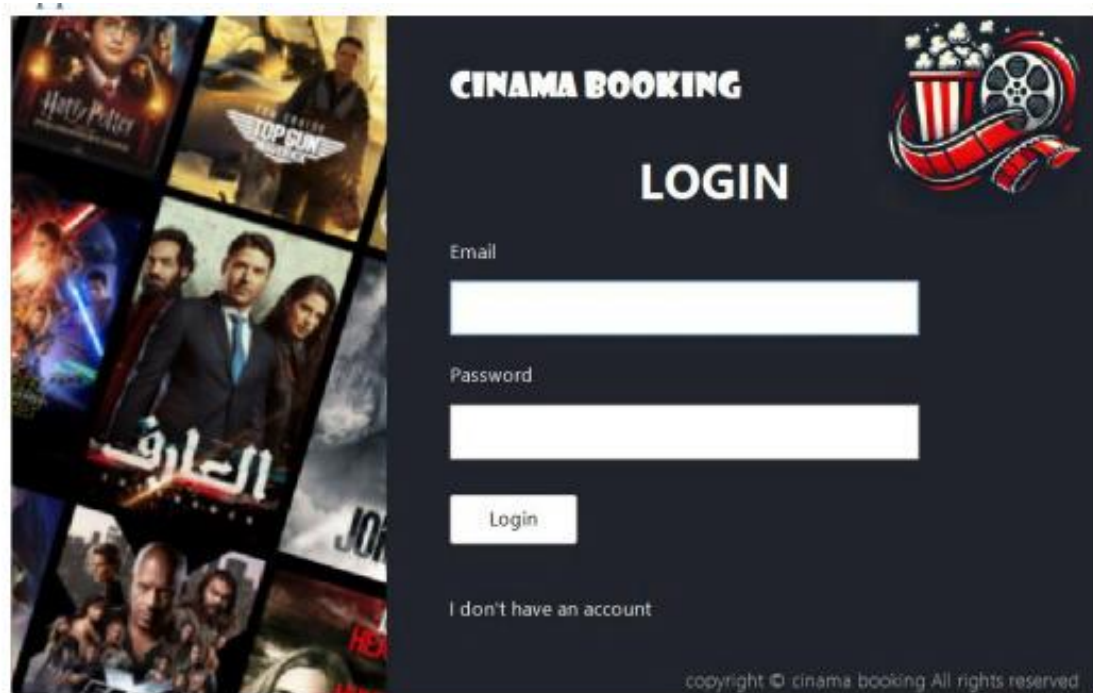
- The cinema website project provided a comprehensive learning experience in both frontend and backend development. It strengthened our understanding of how web technologies work together to create a seamless user experience.
- One of the key accomplishments was successfully integrating a functional booking system that includes movie selection, seat reservation, user authentication, and payment simulation.
- We focused on maintaining a user-friendly interface using modern design principles, which enhances the overall usability and attractiveness of the site.
- Throughout the development process, we faced several challenges such as interface coordination, database connectivity, and handling user inputs securely. These were overcome through collaboration, research, and continuous testing.
- The admin panel feature, including export to CSV, added real-world functionality, giving administrators control over user and ticket data, which simulates how actual cinema systems operate.
- Overall, this project not only improved our technical skills in React, Django, and database management, but also enhanced our teamwork, problem-solving, and project planning abilities.



5. Application's Interface Design:

5.1 Full Application Interfaces and Interactions:

Log In interface



The Login Interface is a central access point for users in the Cinema Booking System. It is designed to authenticate registered users securely and direct them to the appropriate dashboard based on their role (e.g., Admin or User). This form enhances both usability and security, making sure that only verified users

The user enters an email and password.

Input validation is performed (non-empty, proper email format).

Authentication:

Upon clicking the Login button, the system sends the credentials to a DAO class that queries the users table in a MySQL database.

Error Handling:

If credentials are incorrect, a message is displayed: "Invalid email or password."



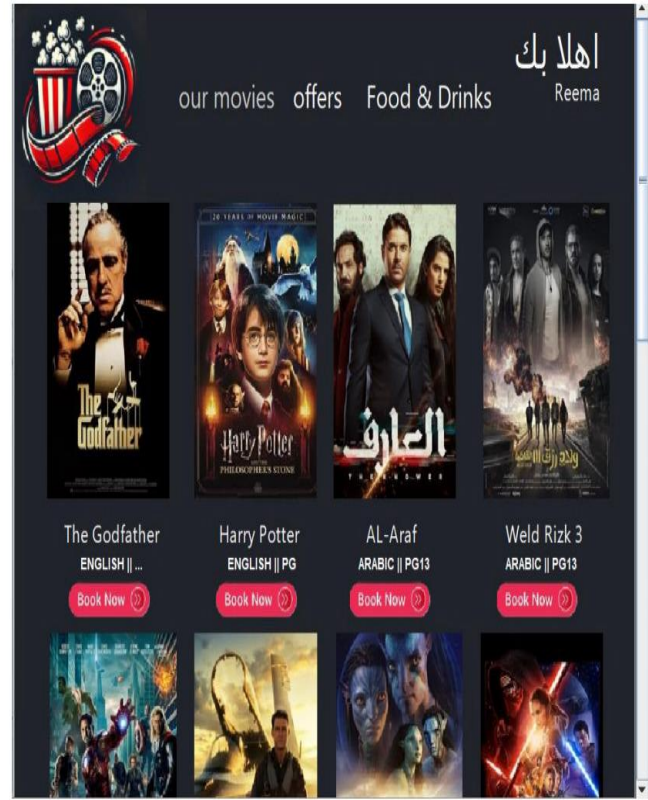
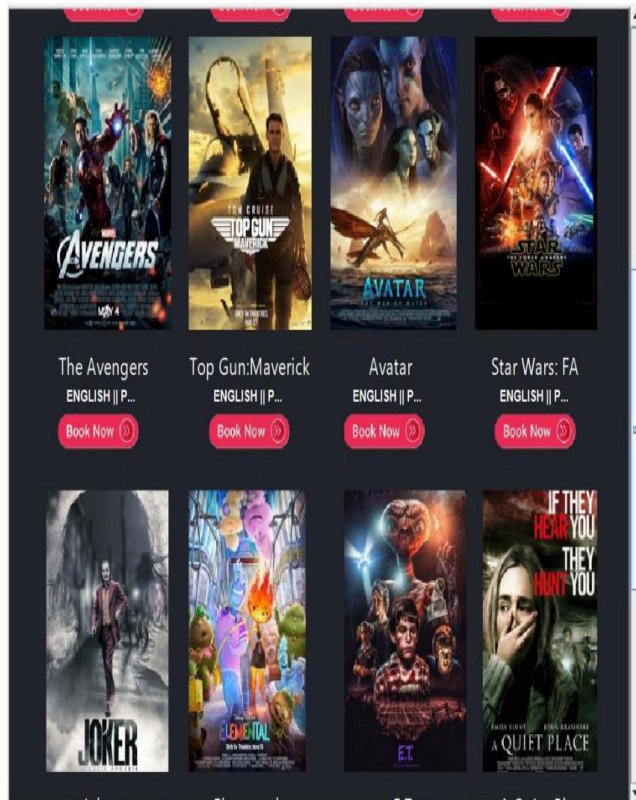
Sign up interface

SIGN UP

First name	Middle name	
Last name	Birthday Date	
	YYYY-MM-DD	
Email	Password	
Phone	Gender	Role
	F	User
SIGN UP		
I've an account		

The Sign Up Interface is part of a cinema booking system that allows new users to create an account by filling out a registration form. This form collects essential personal information and stores it securely in a database. The design is clean, user-friendly, and ensures that users provide all necessary data before gaining access to the system.

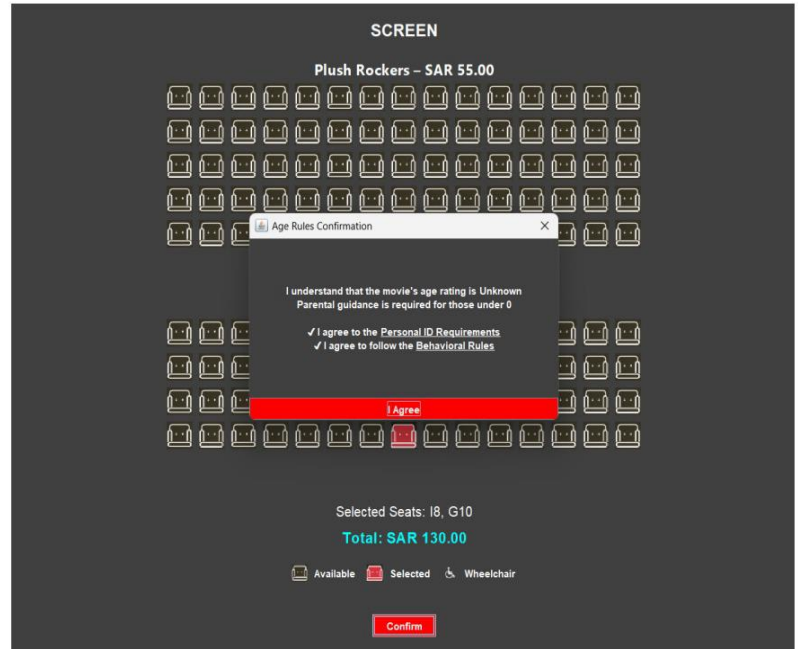
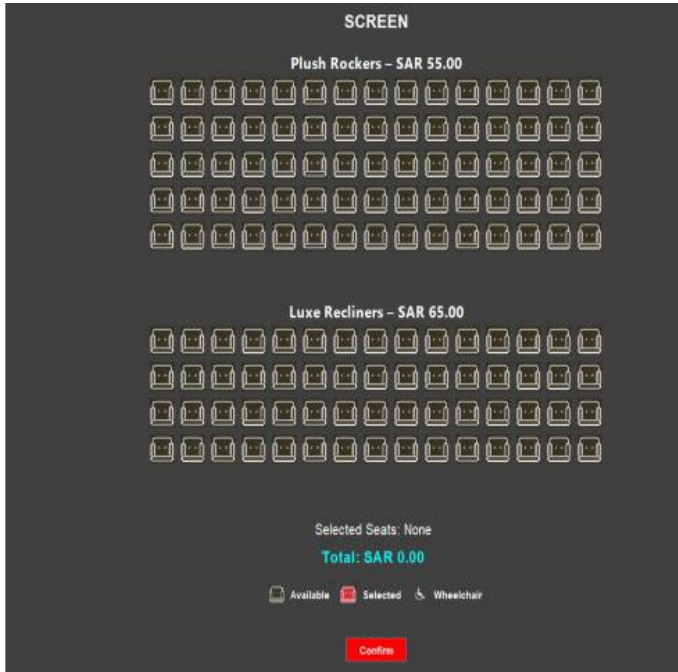
Movie Interface



The Movie Listing Interface is the central display page in the Cinema Booking System that allows users to view available movies, check basic information, and proceed with ticket booking. It provides a dynamic and visually engaging layout that simulates the real experience of browsing through a digital cinema catalog.

This screen is personalized with a greeting ("اهلاً بك Reema") and organized by categories like "Our Movies", "Offers", and "Food & Drinks".

Seat Interface



This module is part of a Cinema Booking System where users can visually select seats for a movie and confirm the booking. The interface includes features like seat pricing, visual indicators for availability, and an age confirmation popup to ensure compliance with viewing restrictions.

Seat Map: A grid layout where each icon represents a seat.

Seat Types with Pricing:

Plush Rockers: SAR 55.00

Luxe Recliners: SAR 65.00

Interactive Selection:

Users click on available seats to select/deselect them.

Selected seats are highlighted in red.

Legends:

Gray: Available

Red: Selected

Wheelchair icon: Accessible seating

Total Calculation: The total cost is dynamically updated based on selected seats.

Confirmation Button: A red "Confirm" button is used to proceed with the booking.

after the user selects a seat and presses

"Confirm", a modal popup appears to enforce age-related rules.

Components:

Title: Age Rules Confirmation

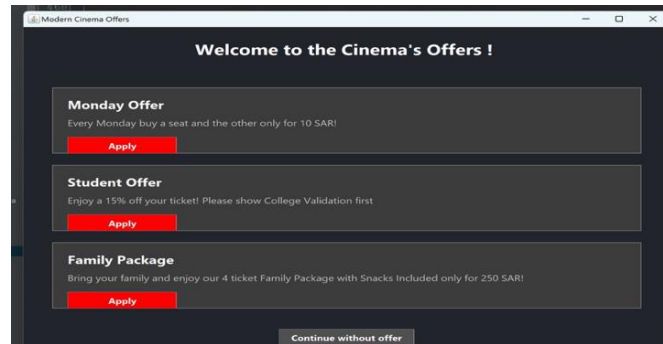
Message:

Informs the user about the movie's age rating.

States that parental guidance is required if the user is under a certain age.



Offers Interface



This interface is part of the Modern Cinema Offers system and is designed to allow users to select from a range of promotional deals before finalizing their ticket purchase. The layout is clean and user-friendly, presenting three clearly defined offers: the Monday Offer, which gives a second ticket for 10 SAR when buying one on a Monday; the Student Offer, which provides a 15% discount with college validation; and the Family Package, offering a bundle of four tickets with snacks for 250 SAR. Each offer includes a bold red “Apply” button, making selection intuitive. Additionally, users who do not wish to apply any offer can proceed directly by clicking the “Continue without offer” button at the bottom. This interface enhances the user experience by presenting value-driven choices in a visually clear and accessible manner.



Food & Drink Interface

FOOD & DRINKS

POPCORN

☐ Regular Popcorn 21 SR
☐ Regular+3.0 SR
☐ Large+4.0 SR

1 +

☐ Caramel Popcorn 22 SR
☐ Regular+3.0 SR
☐ Large+4.0 SR

1 +

☐ Cheese Popcorn 22 SR
☐ Regular+3.0 SR
☐ Large+4.0 SR

1 +

FOOD

☐ NewYorker Pizza Slice 15 SR

1 +

☐ Hotdogs 16 SR

1 +

☐ Burger 18 SR

1 +

☐ Nachos 25 SR

1 +

Drinks

☐ Water 7 SR

1 +

☐ Pepsi 16 SR
☐ Regular+2.0 SR
☐ Large+4.0 SR

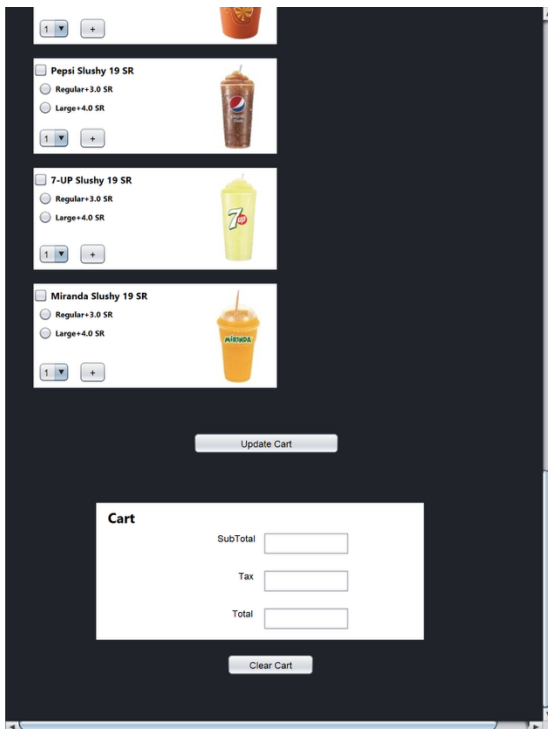
1 +

☐ 7-UP 16 SR
☐ Regular+2.0 SR
☐ Large+4.0 SR

1 +

☐ Miranda 16 SR
☐ Regular+2.0 SR
☐ Large+4.0 SR

1 +



The Food and Drinks GUI is designed to provide a smooth and easy shopping experience. Users can choose from a wide variety of food, popcorn, and drink options, and select the size of drinks or the portion size of popcorn. The quantity of each item can also be specified before adding them to the cart. After an item is added, users can click the “Update Cart” button to refresh and view important details like the subtotal, tax, and total cost. Users can continue to add more items or adjust quantities as needed. If necessary, there’s also an option to clear the entire cart and start over. The design ensures a straightforward, hassle-free shopping process, allowing users to easily manage their cart.

Payment Interface

Seats: 1

Total: 55.0 SAR

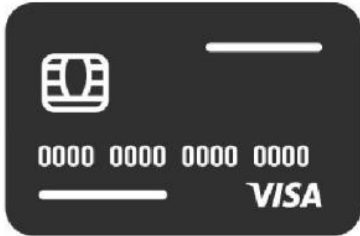


Payments

Credit C... 

Apple P... 

Cash



Card Number

1234567891

Name on The Card

OOPPROJECT

CVV code

244

Expiry d...

06 25

Pay

a simple and clear payment flow that covers both movie and food purchases, while guiding the user through any errors smoothly. After choosing a seat or food item, the user is taken to the payment screen, where the total price is shown along with three payment options: credit card, Apple Pay, or cash.

If the user picks credit card, they'll need to enter their card number, name, CVV, and expiry date. Once everything is entered correctly, a small pop-up confirms the payment was successful, and the system takes the user back to the main screen with the movie listings.

The interface also handles common mistakes to help users avoid issues:

- If they forget to fill out a field, an error message tells them to complete all the fields.
- If the CVV has more than 3 digits, it shows a message saying it must be exactly 3.
- If the card is expired, it alerts the user right away.



Update movie information

Admin Panel - Welcome Admin

Update Movie Information

Movie ID to Replace: New Title:

New Genre:

New Duration (min): New Age Rating:

New Image File Name:

This interface is part of the Admin Panel in a Cinema Booking System. It enables an administrator to update movie details in the system's database. This function is crucial for keeping the movie list current, accurate, and reflecting the most recent changes, such as new posters, genre updates, or age restrictions.

The form provides a simple and effective layout for modifying any movie record based on its unique ID.

Admin fills out the form with new details for a specific movie.

Clicking "Choose Image" opens a file dialog to select a new poster.

Clicking "Update Movie" triggers the following logic:

Validate that all required fields are filled.

Connect to the database using JDBC.

Use an SQL UPDATE query to modify the movie record.

Display a success or error message depending on the result.



5.2 Data Export/Import:

- Import Functionality:

Imports are used not only to bring in necessary libraries for GUI and database interaction, but also to provide the administrator access to retrieve information about users and their bookings. For example, Imports through Scanner in FoodMenu.java and Cart.java, also importing java.sql.* allows the admin to connect to the database and fetch user-related data, such as ticket purchases and personal information. This access supports the system's admin panel functionalities like managing bookings and generating reports.

- Export Functionality:

In the AdminPanel.java file, an Export Tickets to CSV button is implemented. When clicked, it calls the method exportTicketsToCSV(), which gathers the ticket booking data and writes it into a file named tickets_export.csv. This export allows the administrator to save or analyze booking records externally, enabling better data reporting and record keeping.

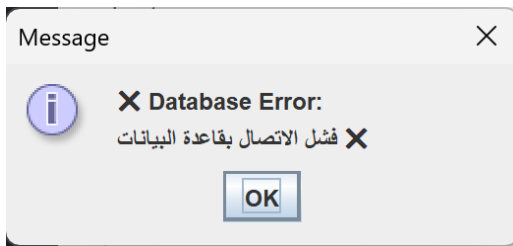
5.3 Error Handling

Error Handling for Missing Selection:



If a user attempts to add an item to the cart without selecting it, the system will detect this missing input. In response, a dialog box or pop-up message will appear on the screen, notifying the user with a clear and concise error message.

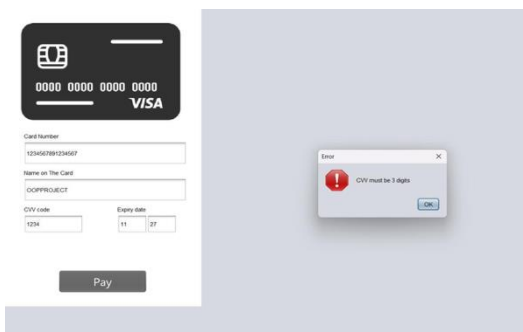
Error handling for Database Connection:



Error Handling for Invalid credentials:

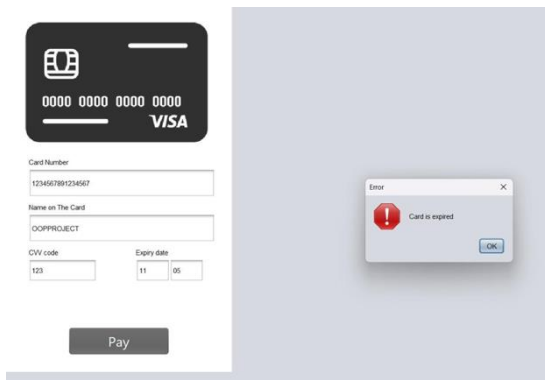
If credentials are incorrect, a message is displayed: "Invalid email or password."

CVV wrong input Error Handling:



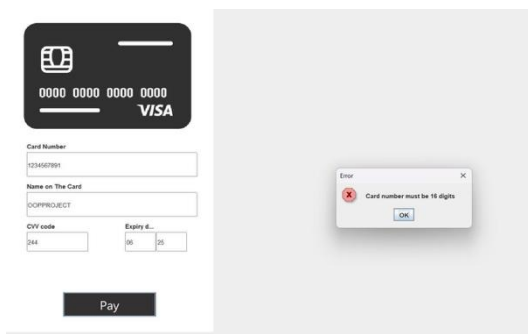
If the CVV has more than 3 digits, it shows a message saying it must be exactly 3.

Expired Card Detection Error Handling:



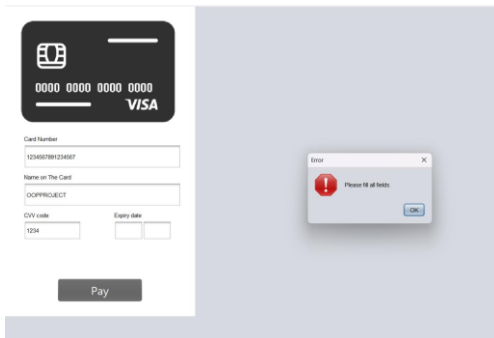
If the entered expiration date indicates a past month or year, the system warns the user that the card is expired.

Wrong number of digits entered Error Handling:



If the entered number of digits does not equal 16 digits, the system warns the user that the card number is wrong.

Missing Field Error Handling:



If user Forgot to fill out a field, an error message tells them to complete all the fields.

Student Offer Validation Error:

If a user selects the student discount offer but does not enter their student ID or college name, an error appears asking for those details.

Monday Offer Error:

If a user selects the Monday offer on any day other than Monday, a warning notifies them that the offer is unavailable today.

Family Offer Validation Error:

When a user selects the family offer without choosing exactly four tickets, an error message is shown requiring them to meet this condition.



6. Attachments:

6.1 Source code

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